ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** vegetative propagation by suckers	Banana is seedless.
2	**(a)** bisexual flower	Has both.
3	**(d)** 46	23 + 23.
4	**(d)** absent	Pregnancy.
5	**(d)** grafting	Joining two varieties.
6	**(d)** unisexual	Separate male and female.
7	**(c)** 46	Same number.
8	**(b)** the 14th day of a 28-day cycle	Mid-cycle.
9	**(d)** vegetative reproduction	New plantlets.
10	**(a)** cross-pollination (or between flowers)	Different flowers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They come from a single parent (1); Genetically identical clones (1).	1+1
12	Formed by mitosis from the parent body (1); No gametes involved (1).	1+1
13	To survive unfavourable conditions (1); Germinate when conditions are right (1).	1+1
14	To attract insects for pollination (1); Help transfer pollen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sperm: small, motile, tail (1); Ovum: large, non-motile, stored food (1); Sperm: DNA delivery; ovum: nutrition for zygote (1).	1+1+1
16	Hen: eggs with shells, external development (1); Humans: viviparous, development in uterus (1); Both internal fertilisation (1).	1+1+1
17	Protect mother and baby (1); Balanced diet for growth (1); Regular check-ups (1).	1+1+1
	OR: Pollen grain germinates on the stigma to form a pollen tube (1); The pollen tube carries the male gamete to the ovule (1); The male gamete fuses with the egg to form a zygote (1).	1+1+1
18	Identical from one zygote (1); Fraternal from two eggs and two sperms (1); Different genes (1).	1+1+1
	OR: Planaria: whole body from pieces (1); Hydra: lost tentacles (1); Starfish: arms (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Gametes formed in testes and ovaries (1); Fertilisation in the oviduct (1); Implantation in the uterus (1); Development with placenta (1); Birth after about 9 months (1).	1+1+1+1+1
	OR: Seed coat (1); Cotyledons (1); Embryo: radicle, plumule (1); Endosperm in maize (1); Germination steps (1).	1+1+1+1+1
20	Vegetative: cutting, layering, grafting (1); Tissue culture: cells in nutrient medium (1); Both give clones (1); Examples (1); Advantages: disease-free, rapid (1).	1+1+1+1+1
	OR: Amoeba: binary fission (1); Yeast: budding (1); Hydra: budding (1); Plasmodium: multiple fission (1); Bread mould: spores (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.

- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** pollen tube	The pollen tube carries the male gametes to the ovule.
2	**(c)** implantation of the embryo	IUD.
3	**(a)** outgrowths that detach	Budding.
4	**(c)** style	To reach the ovule.
5	**(c)** zygote	Fusion of gametes gives the zygote.
6	**(b)** females	Blocking oviducts.
7	**(c)** fragmentation	Breaking into pieces.
8	**(d)** stigma	Pollen lands on the stigma.
9	**(b)** vas deferens	Duct.
10	**(b)** AIDS	Sexually transmitted.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Carries male gametes to the ovule (1); Grows through the style (1).	1+1
12	Sperm carries X or Y (1); Egg always carries X (1).	1+1
13	Many sperms are lost (1); Only one is needed to fertilise (1).	1+1
14	Greater chance of fertilisation (1); Better protection of embryo (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Grafting (1); Fruit quality of the variety is maintained (1); Seeds may not give the same fruit (1).	1+1+1
16	Boys: voice, moustache, sperm production (1); Girls: menstruation, breast development (1); Hormones control these (1).	1+1+1
17	Produce gametes (1); Produce hormones (1); Control secondary sexual characters (1).	1+1+1
	OR: Mitosis: identical cells (1); Meiosis: half chromosomes (1); Variation in sexual reproduction (1).	1+1+1
18	Oestrogen builds the lining (1); Progesterone maintains it (1); Pregnancy hormones prevent menstruation (1).	1+1+1
	OR: Embryo develops on yolk (1); Albumen gives water and protein (1); Shell protects; incubation 21 days (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sepals and petals (1); Anther and filament (1); Stigma, style, ovary (1); Ovules in the ovary (1); Functions (1).	1+1+1+1+1
	OR: Meiosis halves the chromosomes (1); Crossing over (1); Fertilisation restores the number (1); New combinations (1); Natural selection acts (1).	1+1+1+1+1
20	Wind: grass (1); Insects: sunflower (1); Birds: hibiscus (1); Water: Vallisneria (1); Adaptations (1).	1+1+1+1+1
	OR: Resources (1); Health (1); Education (1); Family planning (1); Quality of life (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

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THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Planaria	Cut pieces regrow.
2	**(c)** pollinators	Insects and birds.
3	**(b)** uterus	Womb.
4	**(d)** sexually transmitted infections	Barrier.
5	**(c)** buds	Grow into new plants.
6	**(b)** ovary	After fertilisation.
7	**(a)** amniotic fluid	Shock absorber.
8	**(b)** identical twins	Same genes.
9	**(b)** many spores are produced in a small space	Increase chance.
10	**(d)** integuments of the ovule	Ovule covering.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Health risks to mother and baby (1); Loss of education and opportunity (1).	1+1
12	Supplies food and oxygen (1); Removes carbon dioxide and wastes (1).	1+1
13	Uterus lining thickens (1); Ready for implantation of the embryo (1).	1+1
14	Promotes root growth (1); Helps the cutting establish (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Egg (1); Tadpole with gills (1); Froglet and adult (1).	1+1+1
16	Shortage of food and housing (1); Pressure on resources (1); Family planning (1).	1+1+1
17	Greater variation (1); Better adaptation (1); Improved survival (1).	1+1+1
	OR: Prevents overcrowding (1); Colonise new areas (1); Methods: wind (maple), animals (burr) (1).	1+1+1
18	Spacing of children (1); Health of mother (1); Population control (1).	1+1+1
	OR: Menstruation occurs only in primates (1); Other mammals reabsorb the uterus lining (1); Oestrous: mating at certain periods (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Testosterone (1); Oestrogen (1); Progesterone (1); Menstrual cycle (1); Pregnancy (1).	1+1+1+1+1
	OR: Sperm: head, middle piece, tail (1); Ovum: large with yolk (1); Journey of sperm (1); Fusion forms zygote (1); Block to other sperm (1).	1+1+1+1+1
20	Barrier methods (1); Pills (1); IUDs (1); Surgical (1); Counselling and medical advice (1).	1+1+1+1+1
	OR: Hygiene during puberty (1); STIs: causes and prevention (1); Protection methods (1); Regular check-ups (1); Awareness and counselling (1).	1+1+1+1+1

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